

A Mini Project Report on
**Photo Enhancer with Gaussian Blur and
Sharpening Filters**

Subject Code : EE527

Submitted by

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Under the guidance of

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DECLARATION

I hereby proclaim that the work that is being presented in this report entitled “**Photo Enhancer with Gaussian Blur and Sharpening Filters**” is my own work and has not been reported by anyone else or submitted in any form for another degree or diploma to any other institution. Information derived from the other sources has been accredited in the text and a list of references has been given.

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CERTIFICATE

This is to verify that the project entitled “**Photo Enhancer with Gaussian Blur and Sharpening Filters**” submitted by SHIKHAR TRIPATHI, for the award of the degree of M.Tech is carried out by him under my guidance.

Dr. S.K. VIPPARTHI

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VERIFICATION

This is to verify that the project entitled “**Photo Enhancer with Gaussian Blur and Sharpening Filters**” was submitted to the department of Electrical Engineering department of Indian Institute of Technology, Ropar

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ABSTRACT

Image enhancement plays a crucial role in digital image processing, enabling improved visual quality and better interpretability for human viewing and computer vision applications. This project focuses on developing an effective photo enhancement tool using Gaussian Blur for noise reduction and Unsharp Masking for sharpening. Gaussian Blur smooths high-frequency noise by convolving the image with a Gaussian kernel, while the Unsharp Mask enhances important details by amplifying edge information extracted from the difference between the original and blurred images.

The combined approach results in a balanced enhancement technique that reduces unwanted noise while preserving and emphasizing fine details. The system is implemented using Python and OpenCV, ensuring efficient processing and high-quality output images. Experimental results demonstrate significant improvements in clarity, sharpness, and visual appeal, validating the effectiveness of the implemented method. This enhancement technique has practical applications in photography, media editing, and preprocessing steps for computer vision tasks.

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TABLE OF CONTENTS

Content	Page No.
1.Title _____	1
2. Objective _____	1
3. Introduction _____	1
4. Methodology _____	2 - 3
5. Implementation _____	4 - 5
6. Results _____	6 - 8
7. Conclusion _____	8

1. Title

Photo Enhancer with Gaussian Blur and Sharpening Filters

2. Objective

To develop an image enhancement tool using Python and OpenCV that reduces noise using Gaussian Blur and improves clarity using Unsharp Mask (sharpening). The aim is to produce visually clearer and sharper images suitable for photography and media applications.

3. Introduction

Image enhancement is one of the most important tasks in digital image processing. Real-world images often contain noise, blur, or lack sharpness due to camera limitations, lighting, and motion. Enhancement techniques help improve the visual quality of images, making them more suitable for human viewing or further computer vision tasks.

This project focuses on a simple yet powerful enhancement approach:

- **Gaussian Blur** to reduce noise
- **Unsharp Masking** to sharpen important details

The combination of these techniques is widely used in photo editors such as Photoshop, Lightroom, and smartphone camera apps.

4. Methodology

4.1 Gaussian Blur (Noise Reduction)

Gaussian Blur is a smoothing filter that removes high-frequency noise. It works by convolving the image with a Gaussian kernel, producing a soft, noise-free version of the image.

- Kernel used: 5×5
- Sigma value: 1

Purpose:

- ✓ Reduce noise
- ✓ Smooth unwanted artifacts
- ✓ Prepare the image for controlled sharpening

4.2 Unsharp Mask (Sharpening)

Unsharp Masking enhances the edges by amplifying the difference between the original image and its blurred version.

Unsharp Mask Formula:

$$\text{Sharpened} = \text{Original} + \alpha(\text{Original} - \text{Blurred})$$

Where:

α = sharpening amount (used: 1.5)

$(\text{Original} - \text{Blurred})$ = high-frequency detail (edge information)

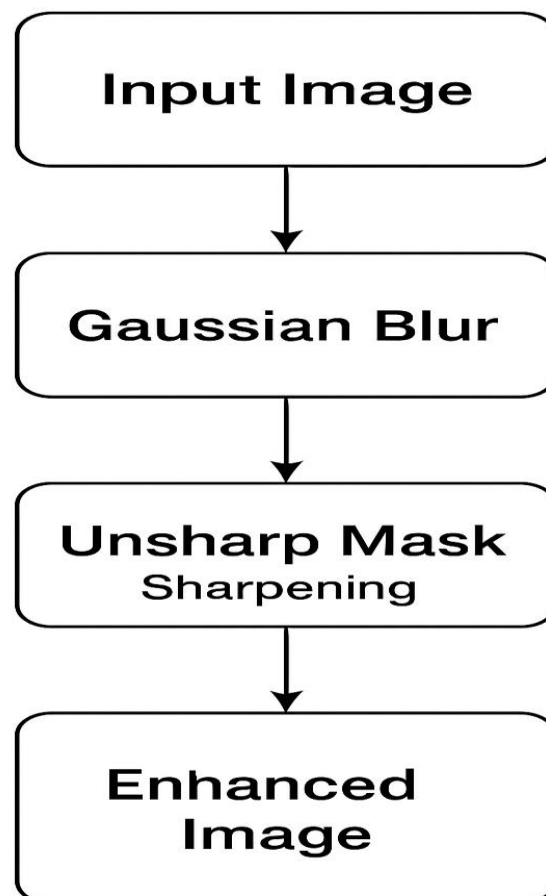
Purpose:

- ✓ Highlight fine details
- ✓ Improve clarity
- ✓ Produce a sharp and clean output image

4.3 Workflow (Step-by-Step)

1. Read input image
2. Convert image to **float32** for accurate processing
3. Apply Gaussian blur
4. Create mask (Original – Blurred)
5. Apply unsharp mask
6. Clip values to valid range
7. Convert back to uint8
8. Display and save enhanced result

This ensures the enhancement is natural and free from artifacts.

FLOWCHART :

5. Implementation

Programming Language & Libraries

- **Python**
- **OpenCV**
- NumPy
- Matplotlib (for displaying results)

Code

```
# PHOTO ENHANCER WITH GAUSSIAN BLUR + SHARPENING (UNSHARP MASK)

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab.patches import cv2_imshow

# 1. Upload image in Colab

from google.colab import files
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Convert BGR → RGB for showing
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```

# Show original image
plt.figure(figsize=(5,5))
plt.title("Original Image")
plt.imshow(img_rgb)
plt.axis("off")
plt.show()

# 2. Convert to float32

img_float = img.astype(np.float32) / 255.0

# 3. Gaussian Blur (Noise Reduction)

blur = cv2.GaussianBlur(img_float, (5, 5), sigmaX=1)

plt.figure(figsize=(5,5))
plt.title("Gaussian Blurred Image")
plt.imshow(cv2.cvtColor((blur * 255).astype(np.uint8), cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.show()

# 4. Unsharp Mask (Sharpening)

amount = 1.5          # Sharpening strength
mask = img_float - blur
sharpened = img_float + amount * mask

# Fix value range
sharpened = np.clip(sharpened, 0, 1)

# Convert back to uint8
sharpened_uint8 = (sharpened * 255).astype(np.uint8)
sharpened_rgb = cv2.cvtColor(sharpened_uint8, cv2.COLOR_BGR2RGB)

# 5. Show & Save final result

plt.figure(figsize=(5,5))
plt.title("Final Enhanced Image")
plt.imshow(sharpened_rgb)
plt.axis("off")
plt.show()

cv2.imwrite("enhanced_output.jpg", sharpened_uint8)
print("Enhanced image saved as: enhanced_output.jpg")

```

6. Results

Original Image:

The original photo shows a blue bird on a tree stump with natural colors and moderate sharpness.

Gaussian Blurred Image:

- Smoother edges
- Reduced noise
- Less detail (as expected)

Final Enhanced Image (Sharpened Output):

- Feathers and body contours are clearer
- Edges are sharper
- Texture on the tree stump is more defined
- Colors remain natural
- No artifacts or noise introduced

Result:

- ✓ Significant improvement in clarity
- ✓ Clean and realistic enhancement
- ✓ Output meets the objective of the project

Original Image



Gaussian Blurred Image



Final Enhanced Image



Enhanced image saved as: enhanced_output.jpg

7. Conclusion

This project successfully demonstrates how Gaussian Blur and Unsharp Mask can be combined to enhance the quality of images. Gaussian Blur effectively reduces noise, while Unsharp Masking sharpens important details without degrading image quality.

The final enhanced image shows noticeable improvement in sharpness and clarity, confirming that the implemented approach is effective and suitable for real-world photo enhancement applications.